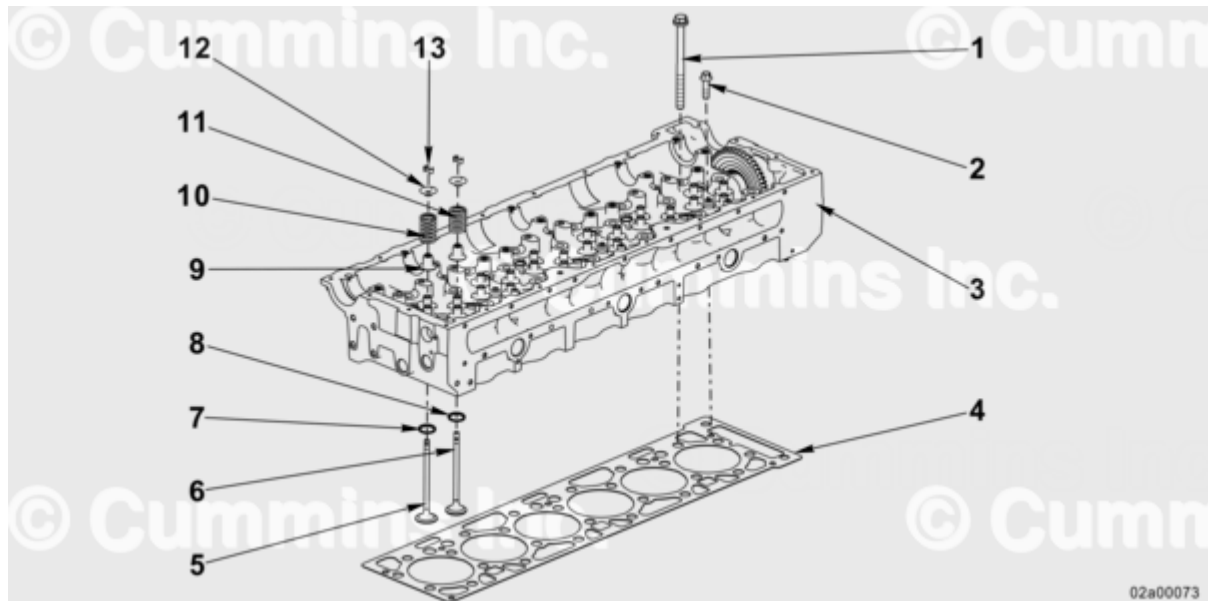


Trainings ▾

Exploded View



Cylinder Head Exploded View

1. Capscrew - quantity 26
2. Capscrew - quantity 2
3. Cylinder head - quantity 1
4. Cylinder head gasket - quantity 1
5. Intake valve - quantity 12
6. Exhaust valve - quantity 12
7. Intake valve insert - quantity 12
8. Exhaust valve insert - quantity 12
9. Valve stem seal - quantity 24
10. Valve spring - quantity 12
11. Valve spring - quantity 12
12. Valve spring retainer - quantity 24
13. Valve collet - quantity 48

Select Service Tools

Recommended Cummins® Service Tools

- Engine lifting fixture, Part Number 3162871
- Oil Passage Protective Plug Kit, Part Number 4919403
- Head holding fixture, Part Number ST-583, or equivalent
- Valve vacuum tester, Part Number 3824278
- Valve Spring Compressor Kit, Part Number 5395402
- Valve Seat Insert Cutting Kit, Part Number 3376405
- Intake seat extractor, Part Number ST-1276-1
- Exhaust seat extractor, Part Number ST-1277-1
- Slide hammer, Part Number 3376617
- Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent
- Chip vacuum, Part Number 3823461, or equivalent
- Crack Detection Kit, Part Number 3375432
- Intake valve seat installation tool, Part Number 3164096
- Exhaust valve seat installation tool, Part Number 3164097

Additional Service Items

- Hoist or hydraulic arm
- Stopwatch
- Soft-faced mallet
- Magnetic tool
- Electric pencil (engraving tool)
- Gasket scraper
- Wire brush, wire wheel (soft), or non-abrasive bead blast.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

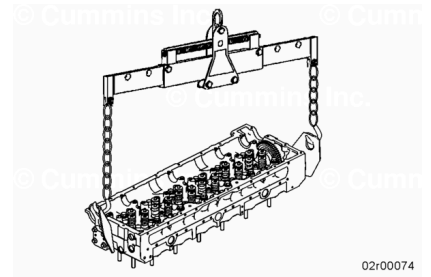
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Remove the air piping from the intake manifold and turbocharger. See equipment manufacturer service information.
- Remove the Exhaust Gas Recirculation (EGR) crossover tube. Refer to Procedure 011-070 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html)
- Remove the EGR cooler. Refer to Procedure 011-019 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-019.html)
- Remove the ignition coils. Refer to Procedure 013-012 in Section 13.
(/qs3/pubsys2/xml/en/procedures/1016/1016-013-012.html)
- Remove the spark plugs. Refer to Procedure 013-016 in Section 13.
(/qs3/pubsys2/xml/en/procedures/1016/1016-013-016-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-011-tr.html)
- Pin the crankshaft. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-088.html)
- Remove the overhead camshaft gear. Refer to Procedure 002-030 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-030.html)
- Remove the camshaft. Refer to Procedure 002-024 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-024.html)
- Remove the internal actuator harness. Refer to Procedure 019-063 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-063.html)
- Remove the engine brakes, if applicable. Refer to Procedure 020-004 in Section 20.
(/qs3/pubsys2/xml/en/procedures/1016/1016-020-004.html)
- Remove the rocker lever assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-009.html)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-001.html)
- Remove the coolant thermostat housing. Refer to Procedure 008-014 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-014.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-033.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-007.html)
- Remove the intake manifold. Refer to Procedure 010-023 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-023.html)
- Remove all the necessary brackets and clamps.

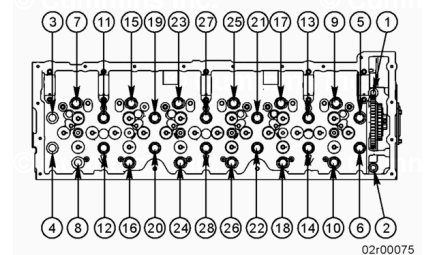
Remove

⚠ WARNING ⚠

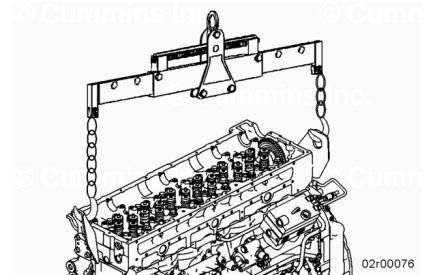
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the cylinder head capscrews in the sequence shown.

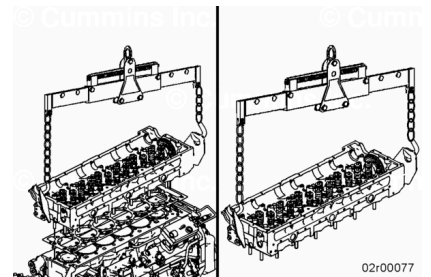


Attach the engine lifting fixture, Part Number 3162871, to the lifting eyes that are bolted to the head.



⚠ CAUTION ⚠

Immediately upon removal of the cylinder head, plugs must be installed into all of the oil passage drillings in both the cylinder head and the cylinder block. The plugs are necessary to prevent debris from entering the lubricating system during the repair. Failure to insert oil passage plugs can result in crankshaft malfunction.



Use a hoist or hydraulic arm to remove the cylinder head.

Remove the cylinder head gasket.

Discard the head gasket.

Reusable plugs are included in the Oil Passage Protective Plug Kit, Part Number 4919403.

Install the plugs in the cylinder block and cylinder head oil drillings.

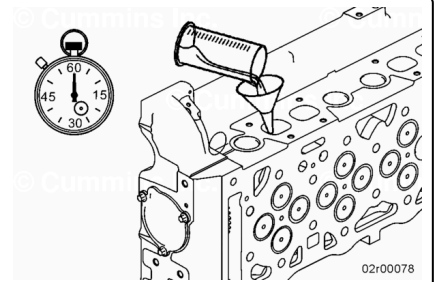
Place cylinder head onto head holding fixture, Part Number ST-583, or equivalent.

Leak Test

Inspect the valves for indications of leaking or burning. If indications of leaking or burning are found, the cylinder head **must** be replaced.

Set the cylinder head down with the exhaust ports facing up. Secure the cylinder head, as it may be unbalanced.

Pour fuel into one of the exhaust ports until it is full. Set the container of fuel down, and start a timer.

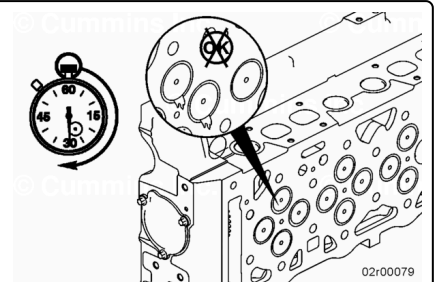


If a drop of fuel drips or runs down the face of the cylinder head within 30 seconds, the exhaust valves and the seats are **not** sealing properly. The cylinder head **must** be replaced.

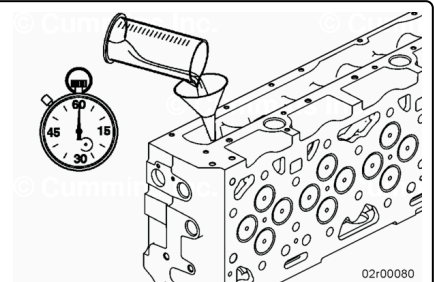
If a drop of fuel has **not** run down the face of the cylinder head in 30 seconds, the exhaust valves are acceptable.

Repeat the process for all six cylinders.

Flip the cylinder head over and set it down with the intake ports facing up. Secure the cylinder head to prevent it from tipping over, as it may be unbalanced.



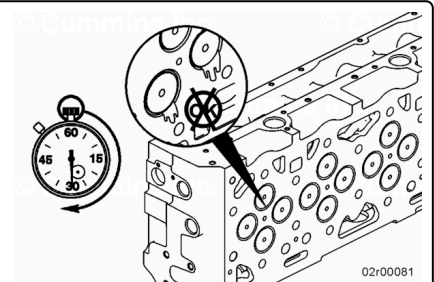
Pour fuel into one of the intake ports until it is full. Set the container of fuel down, and start the timer.



If a drop of fuel drips or runs down the face of the cylinder head within 30 seconds, the intake valves and the seats are **not** sealing properly. The cylinder head **must** be replaced.

If a drop of fuel has **not** run down the face of the cylinder head in 30 seconds, the intake valves are acceptable.

If any of the intake or exhaust valves do **not** pass the test, the cylinder head **must** be replaced.

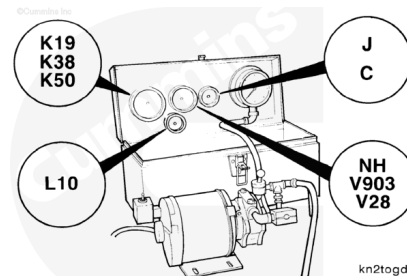


Vacuum Test

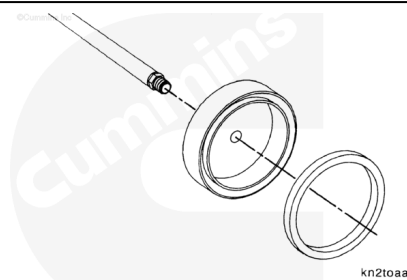
Use the valve vacuum tester, Part Number 3824278, to inspect the seal between the valve and the valve seat.

See the Leak Test section of this procedure for the recommended procedure to check used cylinder heads.

The valve vacuum tester can be used to test all Cummins® engine models.



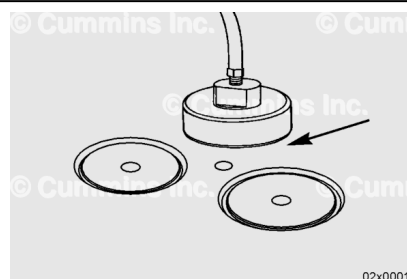
Install the seal ring and vacuum cup to the vacuum line hose.



The valves and valve seats **must** be clean and dry when vacuum testing.

Cover the valve with the cup and seal.

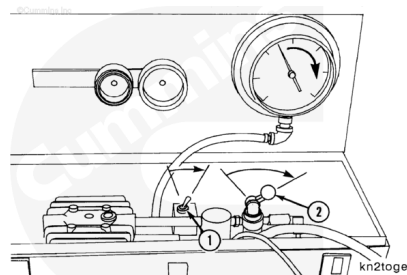
The seal **must** make a tight contact on the cylinder head around the valve.



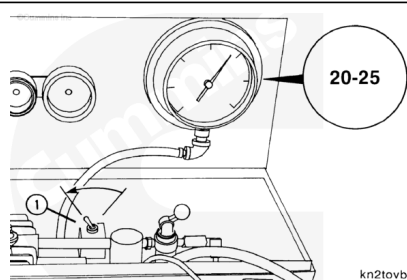
Move the toggle switch (1) to the ON position.

Turn the vacuum control valve (2) to the OPEN position.

The vacuum control valve is in the OPEN position if the vacuum gauge needle moves **clockwise**.



Operate the vacuum pump until the gauge indicates the specified vacuum.



Valve to Valve Seat Vacuum

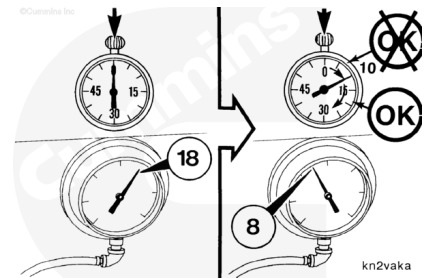
mm-hg		in-hg
508	MIN	20
685	MAX	27

Turn the toggle switch to the OFF position.

Use a stopwatch and start timing when the needle on the gauge indicates 457 mm-hg [18 in-hg].

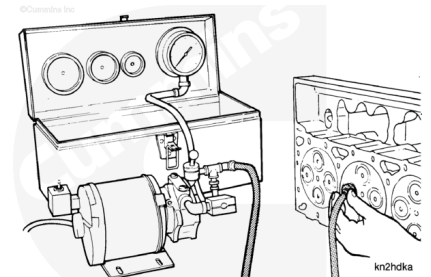
Stop timing when the needle on the gauge indicates 203 mm-hg [8 in-hg].

The elapsed time for the needle to move between the specified gauge readings **must** be 10 seconds or more.



If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to be sure the equipment is operating properly.
- Use a soft-faced mallet to lightly tap the valve head to make sure the valve is seated. Repeat the test.
- Apply a thin layer of grease on the outside circumferences of the insert and the valve head. Repeat the test. The grease pattern will show the point of leakage.



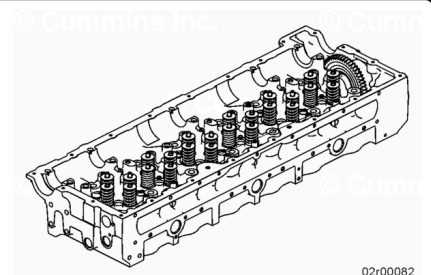
A break in the grease seal pattern will indicate leakage between the valves and valve seat or the valve seat insert and the cylinder head.

If a leak is found, replace the cylinder head.

Disassemble

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Place the cylinder head combustion face down on a clean, smooth, and level work surface that is sturdy enough to support the weight of the cylinder head.

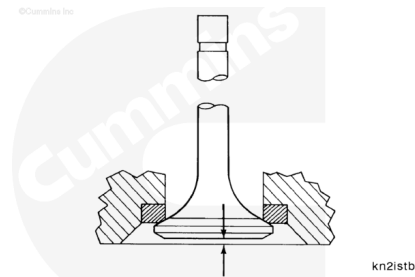
Note : This section shall **only** be used when necessary and is **not** needed in every situation.

Prior to disassembly, measure and record the valve recess.

Verify the measurement tool is flat and flush on the machined cylinder head surface. Measure the flat surface of the valve, avoiding the scalloped areas.

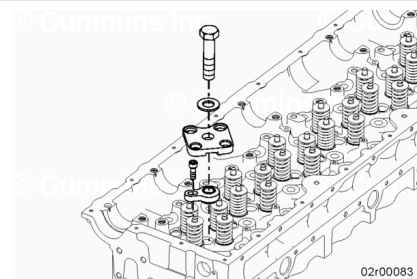
Operational/Reuse Valve Recession		
mm		in
3.8	MAX	0.150

If the valve recess is outside the specifications, replace the valve. If the valve recess is still outside the specifications, check the valve seat insert.



⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.



⚠ CAUTION ⚠

Do not use an impact tool for any part of this procedure. Doing so can damage the tool and/or the engine.

Use the Valve Spring Compressor Kit, Part Number 5395402, to compress the valve springs.

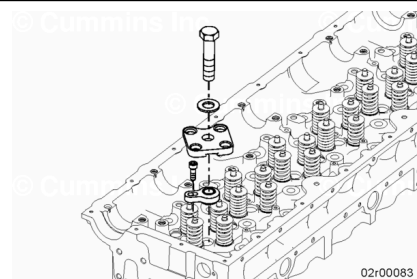
Position the replacer screw pilot to the injector bore and secure with the capscrew in the Valve Spring Compressor Kit, Part Number 5395402.

Position the compressor plate over the valve spring. Put the compressor plate at the head of the exhaust valve and intake valve. Make sure the thin side of the plate on the two exhaust valves, and the thick side of the plate on the two intake valves.

Install the washer and the heavy screw. Screw the heavy screw into the screw pilot.

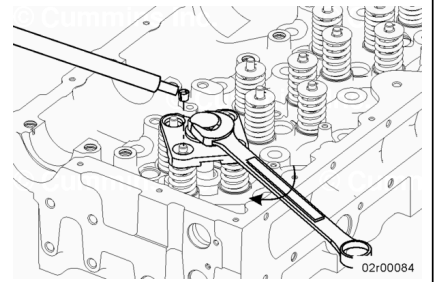
Tighten the heavy screw.

Turn the heavy screw **clockwise** until the valve springs start to compress.



⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.



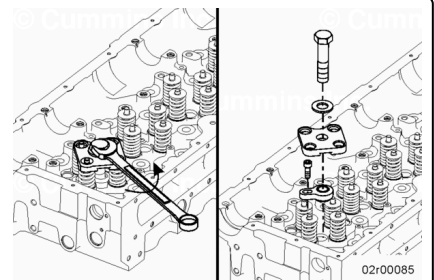
Continue turning the heavy screw **clockwise** until the valve collets can be removed with a magnetic tool.

Remove and discard the valve collets.

Carefully turn the heavy screw **counterclockwise** until the tension is released from the valve springs.

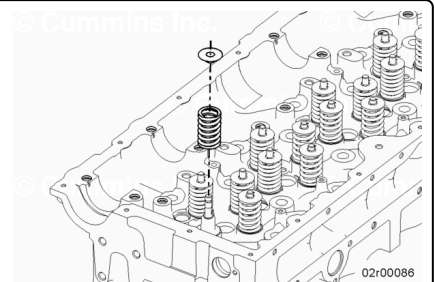
Remove the heavy screw and compressor plate from the screw pilot.

Remove the screw pilot.

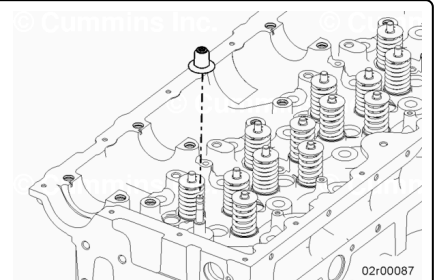


Remove the valve spring retainer.

Remove the valve spring.



Remove the valve seal jacket.

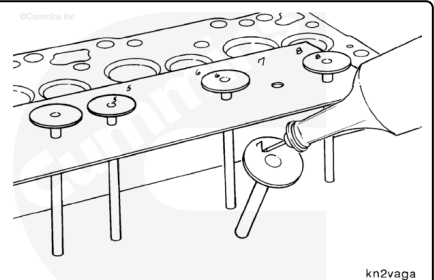


With the use of a lifting device, roll the head up on its side to gain access to the valves.

Remove the valves.

Mark the valves with an electric pencil (engraving tool) for location as they are removed.

The intake and exhaust valves are **not** interchangeable due to being differing sizes.



Valve Seat Insert

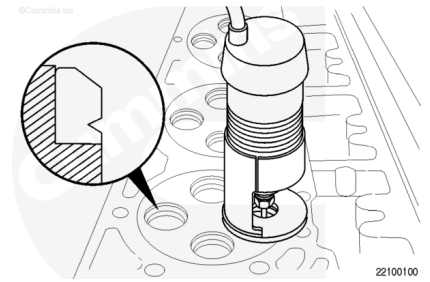
Prior to the removal of the valve seat inserts, see Clean and Inspect for Reuse section in this procedure, which helps to determine if a valve seat insert needs to be replaced.

If required, remove the valve seat insert.

Inspect the valve insert-to-cylinder head contact area. A sufficient groove for the remover **must** exist.

If the valve seat insert groove area is sufficient, proceed to the next step.

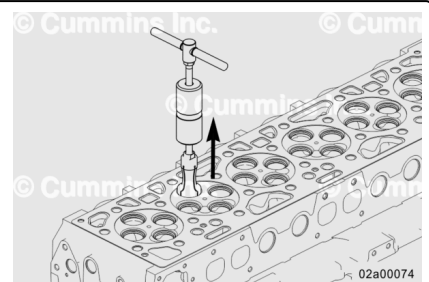
If the valve seat insert groove area is **not** sufficient, use the Valve Seat Insert Cutting Kit, Part Number 3376405, to create a sufficient groove.



Use the required tool to remove the intake and exhaust valve seat insert. Use intake seat extractor, Part Number ST-1276-1, exhaust seat extractor, Part Number ST-1277-1, and slide hammer, Part Number 3376617.

Install the required extractor into the valve seat insert and connect the slide hammer.

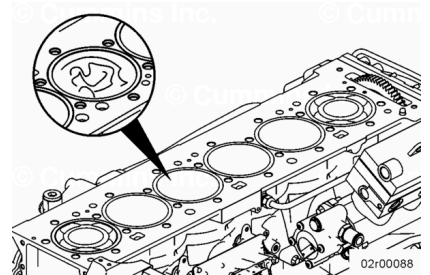
Use the slide hammer to remove the valve seat insert.



Clean and Inspect for Reuse

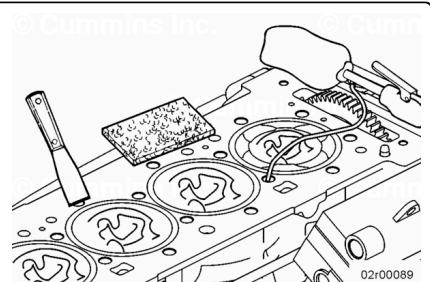
Place clean lint-free rags above the pistons to prevent debris from falling into the engine.

Plug or cover the coolant and oil passages in the cylinder block deck.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a gasket scraper to clean the cylinder block deck surface.

Use Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, and solvent to remove any residual gasket material from the cylinder block deck surface.

Use a chip vacuum, Part Number 3823461, or equivalent, to make sure the cylinder head capscrew holes are clean and free of debris, oil, and coolant.

Do **not** use power tools to clean the cylinder block head deck.

Use the following procedure to inspect the cylinder block head deck. Refer to Procedure 001-026 in Section 1.

(/qs3/pubsys2/xml/en/procedures/1016/1016-001-026.html)

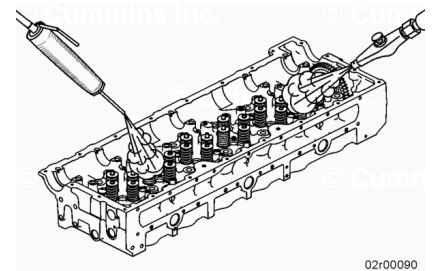
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Steam clean the cylinder head.

Dry with compressed air.

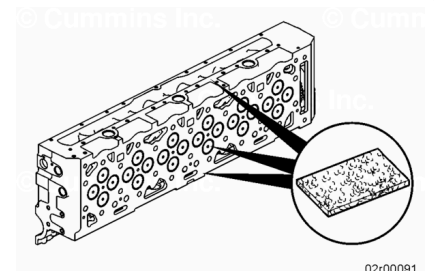
Use compressed air to blow out all capscrew holes.



02r00090

Use Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, and solvent to clean the cylinder head combustion face.

Clean the exhaust manifold and intake manifold gasket surfaces.



02r00091

⚠ CAUTION ⚠

Do not use caustic or acid solutions to clean the cylinder head capscrews to avoid damage to cylinder head capscrews.

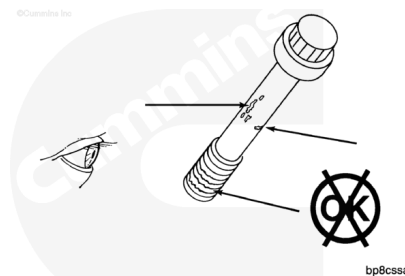
Clean the cylinder head capscrews.

Clean the capscrews thoroughly with a wire brush, a wire wheel (soft), or a non-abrasive bead blast to remove deposits from the shank and threads.

Inspect the capscrews for the following:

- Damaged threads
- Rust or corrosion caused by pitting
- Nicks, bent, stretched, or galled.

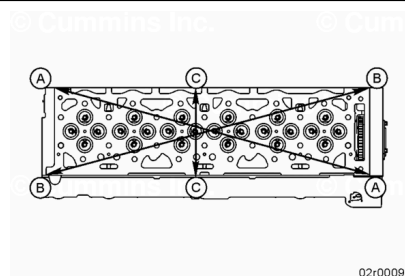
The capscrew **must** be replaced if it has any of the listed damages.



Measure the flatness of the cylinder head combustion face surface as follows:

- A to A and B to B (corner to corner)
- C to C (across combustion face).

Dimensions for C to C **must** be checked at multiple locations from front to rear of the cylinder head.



Cylinder Head Flatness			
	mm		in
A to A	0.15	MAX	0.006
B to B	0.15	MAX	0.006
C to C	0.15	MAX	0.006

If the cylinder head is pitted, grooved, or is **not** within specifications, the cylinder head **must** be replaced.

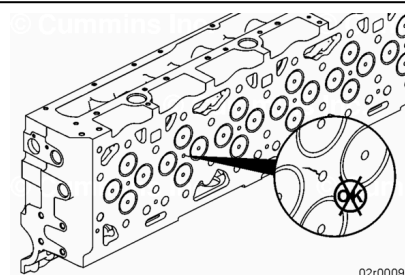
Cylinder Head Cracks - Reuse Guidelines

With the cylinder head cleaned, re-inspect the cylinder head for signs of cracks, fretting, and discoloration that will prohibit reuse.

To help identify cracks in the cylinder head, use the Crack Detection Kit, Part Number 3375432.

Pay close attention to areas of the cylinder head that include:

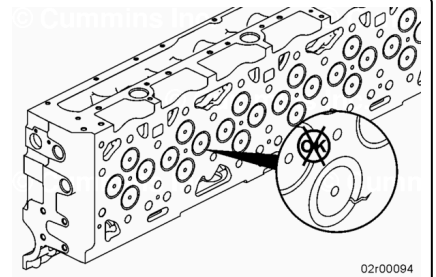
- Injector bore
- Combustion face
- Valve seals.



⚠ CAUTION ⚠

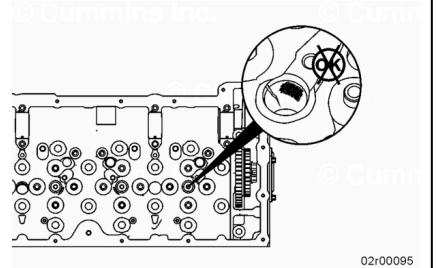
Failure to replace the cylinder head for a crack that extends into or through the valve seat bore will result in a valve seat insert falling out. Engine damage will result.

- If any cracks are found, the cylinder head **must** be replaced.



Inspect the injector bores for wear, cracks, other damage, or mis-drilled passages.

The cylinder head can **not** be used if wear, cracks, other damage, or mis-drilled passages are detected in the injector bores.



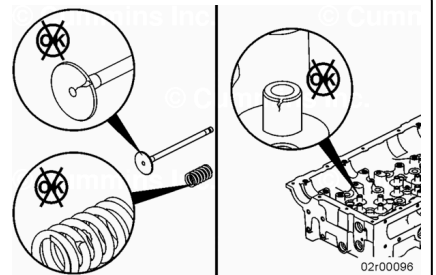
Inspect the valves and valve springs for cracks, bent or broken valve stems, or other damage.

If any damage is found, the valves and valve springs **must** be replaced.

Inspect the valves for indications of leakage or burning.

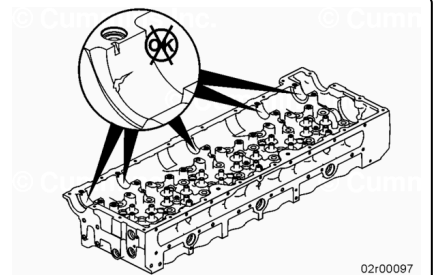
If any of the following damage is found, the cylinder head **must** be rebuilt or replaced.

- Cracks
- Leaks
- Burnt valves
- Other damage.



Clean the camshaft bearing saddles. Make sure to inspect the camshaft bearing saddle areas for cracks, fretting, and signs of discoloration.

If any cracks are found, the cylinder head **must** be replaced.

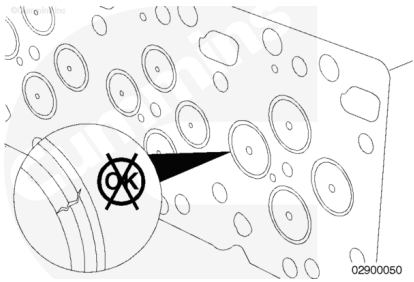


Inspect the valve seat inserts for cracks or burn spots.

If the valve seat inserts are cracked or burned, the valve seat inserts **must** be replaced.

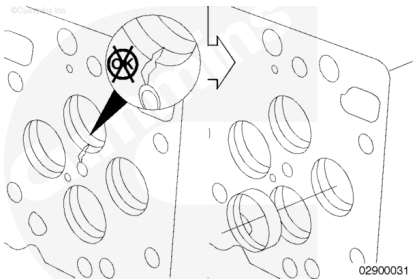
When the valve seat inserts are replaced, inspect the valve seat status together. See Clean and Inspect for Reuse section in this procedure.

If the valve seat has dropped out and caused damage to the cylinder head combustion face, the cylinder head **must** be replaced.



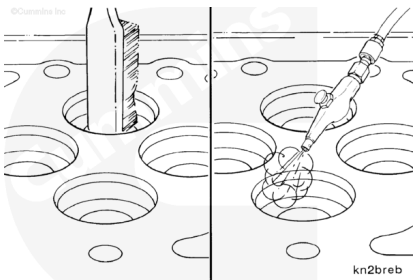
If a crack extends into or through the valve seat bore, the cylinder head **must** be replaced.

Use Crack Detection Kit, Part Number 3375432, to help identify cylinder head cracks.

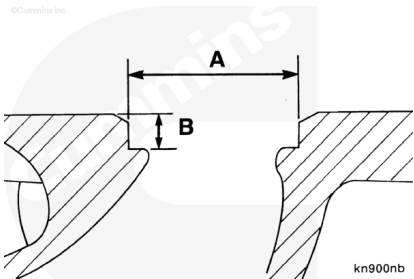


Use a wire brush and solvent to clean the deposits from the valve seat insert bores if the valve seat inserts need to be removed.

Dry with compressed air.



If the valve seat insert is removed in the Disassemble section, measure the inside diameter of the valve seat insert bore (A) and the valve seat insert bore depth (B) in the cylinder head.



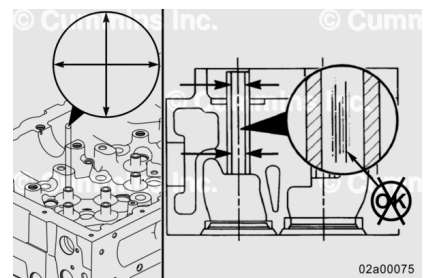
(A) Valve Seat Insert Bore Inside Diameter (Standard Insert)			
	mm		in
Intake valves	46.813	MAX	1.843
Exhaust valves	45.848	MAX	1.805

(B) Valve Seat Insert Bore Depth (Standard Insert)			
	mm		in
Intake valves	8.74	MAX	0.344
Exhaust valves	9.89	MAX	0.389

If the inside circumference of the valve seat insert bore is worn, the cylinder head **must** be replaced.

Inspect the intake valve bore and exhaust valve bore for wear, cracks, other damage, or mis-drilled passages.

Use a small-bore gauge to measure the inside diameter of the intake valve bores or exhaust valve bores at locations 13 mm [0.51 in] from each end and at the center as shown.



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Intake/Exhaust Valve Bore Diameter

mm		in
9.539	MIN	0.376
9.569	MAX	0.377

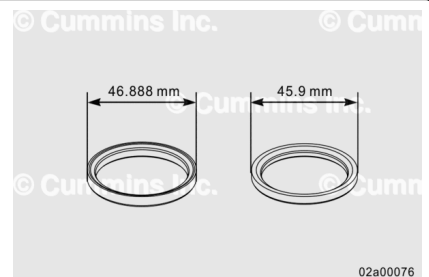
If the intake valve bore or exhaust valve bore diameter is worn larger than the maximum specified, the cylinder head **must** be replaced.

Assemble

Valve Seat

When installing the valve seat inserts, the exhaust and intake valve seat inserts are **not** the same valve seat diameter. Intake valve is bigger than exhaust valve.

- Intake valve seat insert diameter is 46.888 mm [1.846 in].
- Exhaust valve seat insert diameter is 45.9 mm [1.807 in].



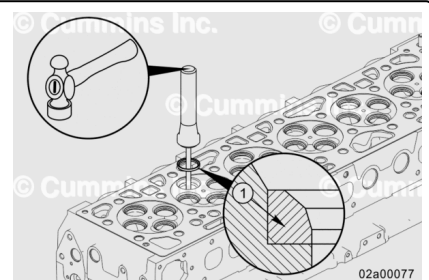
02a00076

The valve seat insert chamfer (1) **must** be installed toward the bottom of the valve seat insert bore.

Install the valve seat insert over the valve seat insert bore in the cylinder head.

Use intake valve seat installation tool, Part Number 3164096, or exhaust valve seat installation tool, Part Number 3164097, to drive the valve seat insert into the valve seat insert bore.

Verify the insert is at the bottom of the valve seat insert bore.



02a00077

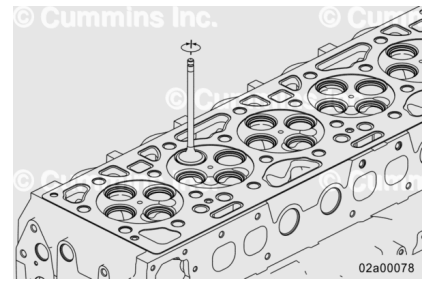
If new valve seat insert/valve is installed and/or the valve leakage is above specification, the valve seat insert/valve can be lapped.

Lubricate the stems with Society of Automotive Engineers (SAE) 15W-40 engine oil before installing the valves.

Apply a thin and even coating of fine lapping compound on the valve.

Use a power or hand suction lapping tool to provide pressure in the center of the valve.

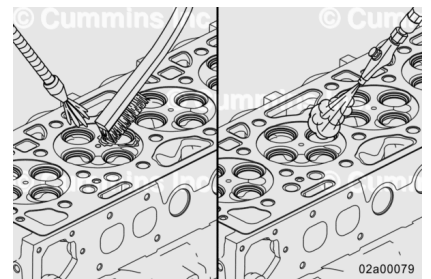
Rotate the valve backward and forward. Continue lapping until the compound shows a continuous contact pattern on both the valve seat insert and the valve.



⚠ CAUTION ⚠

Lapping compound is an abrasive material. Damage will result if the cylinder head, valves, and valve seats are not cleaned thoroughly.

Clean the lapping compound from the parts.

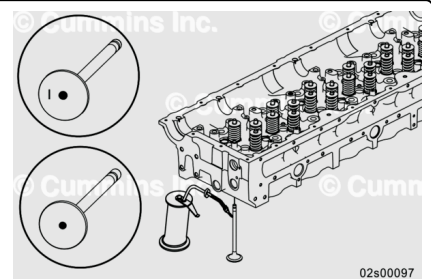


Use clean engine oil to lubricate the valve stems.

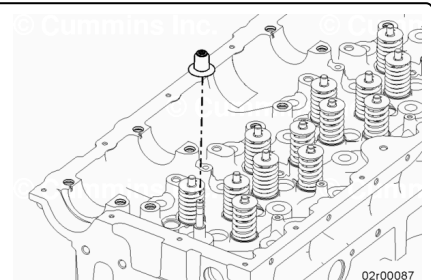
Install the valves.

After the valves are installed, place the cylinder head on a flat surface that will **not** damage the cylinder head surface.

Note : Distinguish the intake valve and exhaust valve and spring. Intake valve: Bottom surface pin with "I" mark and white paint. Exhaust valve: Bottom surface pin with blue paint.



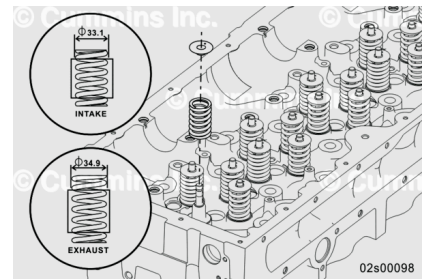
Install the valve seal jacket.



Install the valve springs.

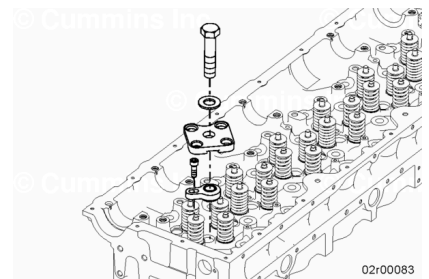
Install the valve spring retainers.

Note : Distinguish the intake valve and exhaust valve and spring. Intake valve spring: White paint, outside diameter (33.1 mm). Exhaust valve spring: Green paint, outside diameter (34.9 mm).



⚠ CAUTION ⚠

**Do not use an impact tool for any part of this procedure.
Doing so can damage the service tool and/or the engine.**



Use the four-valve spring compressor, from Valve Spring Compressor Kit, Part Number 5395402, to compress the valve springs.

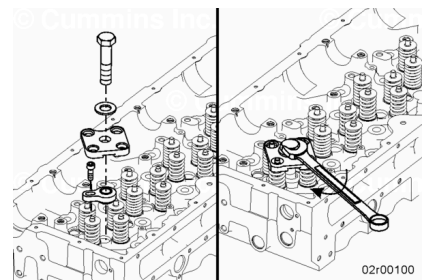
Position the replacer screw pilot to the injector bore and secure with the capscrew in the Valve Spring Compressor Kit, Part Number 5395402.

Position the compressor plate over the valve spring. Put the compressor plate at the head of the exhaust valve and intake valve. Make sure the thin side of the plate on the two exhaust valves, and the thick side of the plate on the two intake valves.

Install the washer and the heavy screw. Screw the heavy screw into the screw pilot.

Tighten the heavy screw.

Turn the heavy screw **clockwise** until the valve springs start to compress.



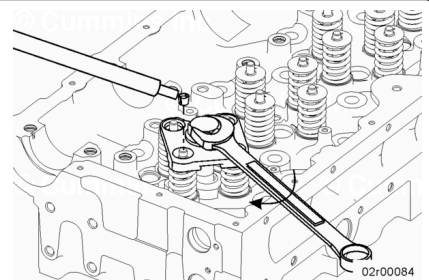
Always use new collets when rebuilding the cylinder head.

Install the new valve spring retainer collets.

Inspect the collets for proper seating and installation.

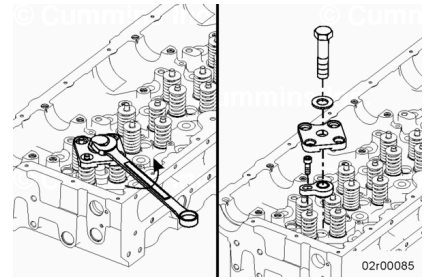
Once the cylinder head is assembled, it will be necessary to run the vacuum test to assure good valve to seat sealing.

See the Vacuum Test section in this procedure.



⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.



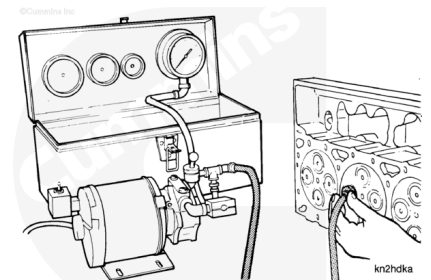
Carefully turn the heavy screw **counterclockwise** until the tension is released from the valve springs.

Remove the heavy screw and compressor plate from the screw pilot.

Remove the screw pilot.

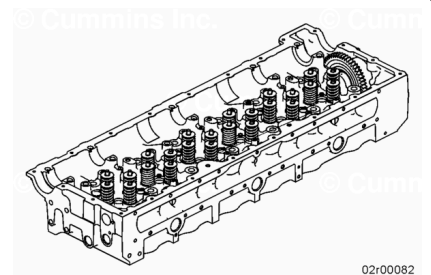
Repeat the above steps for each cylinder as needed.

Before installing the cylinder head, vacuum test the cylinder head again. See the information above in the Vacuum Test section of this procedure.

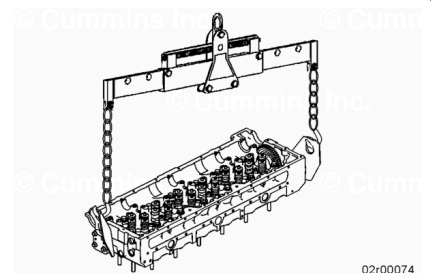


Install

Remove all cylinder head packaging if present.

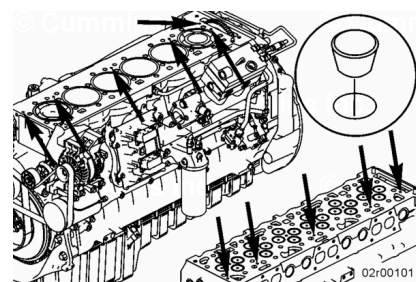


Attach the engine lifting fixture, Part Number 3162871, to the lifting eyes that are bolted to the head.



⚠ CAUTION ⚠

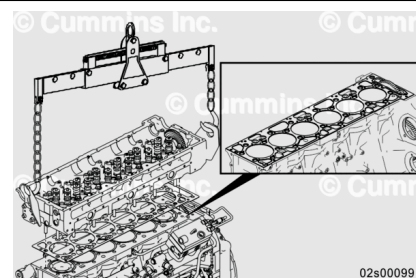
Before the cylinder head is assembled onto the block, all oil passage plugs must be removed. Failure to remove the oil plugs will result in extensive engine damage.



Remove the oil plugs and/or tape from the cylinder head and the cylinder block.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not drop the cylinder head on the cylinder head gasket. The gasket material can be damaged.

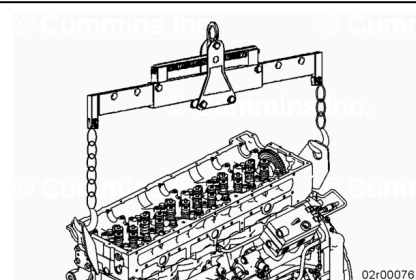
Install a new cylinder head gasket.

Place the cylinder head gasket on the top surface of the cylinder block through the locating pin of the cylinder head, cylinder head gasket locating pin hole waist hole forward.

Check whether the cylinder head gasket is closely connected with the top surface of the cylinder block, and ensure that all water holes, bolt holes, oil holes, and through holes are connected.

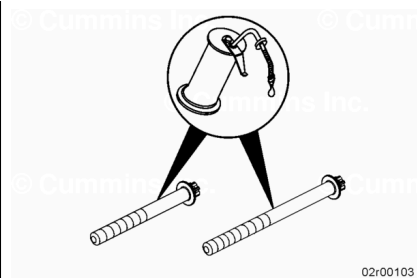
Use a hoist or hydraulic arm and install the cylinder head.

Remove the engine lifting fixture, Part Number 3162871, from the lifting eyes that are bolted to the head.



Use clean lubricating engine oil to lightly coat the cylinder head capscrew threads and bottom of the flange.

Allow the excess oil to drain from the capscrew threads.

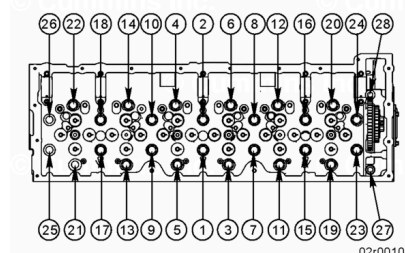


⚠ CAUTION ⚠

Cylinder head capscrews must be installed in the correct location or engine damage may result.

⚠ CAUTION ⚠

Capscrew torque specifications vary according to capscrew length and location. Failure to apply the correct torque value or follow the correct torque sequence may result in engine damage.



Tighten all the capscrews in the sequence shown to the specified values.

Torque Value:

Capscrews 1 through 26 **only**

1. 135 n•m [100 ft-lb]
2. Loosen fully.
3. 135 n•m [100 ft-lb]
4. 200 n•m [148 ft-lb]
5. Turn 120 degrees.

Torque Value:

Capscrews 27 and 28 113 n•m [83 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install all the necessary brackets and clamps.

- Install the camshaft. Refer to Procedure 002-024 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-024.html)
- Install the intake manifold. Refer to Procedure 010-023 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-023.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-033.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-007.html)
- Install the coolant thermostat housing. Refer to Procedure 008-014 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-014.html)
- Install the crossheads. Refer to Procedure 002-001 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-001.html)
- Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-009.html)
- Install the engine brakes, if applicable. Refer to Procedure 020-004 in Section 20.
(/qs3/pubsys2/xml/en/procedures/1016/1016-020-004.html)
- Set the valve lash. Refer to Procedure 003-004 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-004-tr.html)
- Install the overhead camshaft gear. Refer to Procedure 002-030 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-030.html)
- Install the internal actuator harness. Refer to Procedure 019-063 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-063.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-011-tr.html)
- Install the spark plugs. Refer to Procedure 013-016 in Section 13.
(/qs3/pubsys2/xml/en/procedures/1016/1016-013-016-tr.html)
- Install the ignition coils. Refer to Procedure 013-012 in Section 13.
(/qs3/pubsys2/xml/en/procedures/1016/1016-013-012.html)
- Install the EGR cooler. Refer to Procedure 011-019 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-019.html)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html)
- Install the air piping to the intake manifold and turbocharger. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks.

